Yuan Xue

Senior Bioinformatics Data Scientist

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Employment

ClearNote Health — San Mateo, California, USA

Senior Bioinformatics Data Scientist — *Jan 2022 – Present*

- Developed and published (co-first author, **Communications Biology 2025**) a novel cfDNA classifier that localizes tumor tissue-of-origin across 5 cancer types at 85.2% accuracy via multinomial elastic-net logistic regression on 5hmC signatures over a 2,687-subject cohort; pending patent.
- Built production bioinformatics and ML pipelines (Nextflow, Python, PyTorch, scikit-learn) for cancer prediction, tumor-fraction quantification, and genomic feature engineering from low-pass WGS and 5hmC-enriched cfDNA fragmentomics, scaling to thousands of clinical samples.
- Led cfDNA genomics assay optimization in early discovery and development (experimental design, troubleshooting, statistical analysis) across a cross-functional wet-lab and computational team, cutting sample-to-sample variation in plasma protein and DNA quantification by 4X and 2X; enabled its transition into analytical validation of the Avantect Pancreatic Cancer Test (co-author, **J. Mol. Diagnostics 2024**).
- Led sequencing-platform feasibility studies benchmarking base quality (Q30), GC bias, error and duplication rates, coverage uniformity, and alignment quality across 4 commercial sequencing platforms, driving platform selection and cost-efficiency for liquid-biopsy applications.
- Led adoption of internal control standards and engineered automated QC analytics (Nextflow, Snowflake/dbt), enabling process-efficiency tracking and reproducible quality gates across the clinical pipeline; pending patent.
- Contributed 2 peer-reviewed publications and 4 conference presentations (AACR, ESMO, ASHG, CSHL).

Research Experience

Stanford University - Quake Lab — Stanford, California, USA

Doctoral candidate of Bioengineering — *September 2015 – December 2021*

- Developed single-cell transcriptomic methods to profile *Toxoplasma gondii* and infected host cells. Discovered novel antigenic variation and its regulatory transcription factor, and assessed evolutionary concordance of *Toxoplasma* asexual life-cycle with *Plasmodium*. Built an interactive browser for community data visualization. Published in **eLife**.
- Led single-cell profiling of murine macrophage response to persistent *Salmonella typhimurium* infection. Performed functional pathway analysis (GSEA, KEGG, Reactome) linking metabolic and immune pathways to infection response, and identified a novel Ace+ macrophage cell type mediating intracellular bacterial persistence. Published in **Science Advances**.
- Developed biochemical and genomics assays to profile replication fidelity and kinetics of DNA polymerases across evolutionary origins (psychrophilic, mesophilic, thermophilic). Identified a novel cold-active polymerase from *Psychromonas ingrahamii*. Published in **Journal of Biological Chemistry**.

Education

Stanford University — Stanford, California, U.S.A.

Ph.D. and M.S. in Bioengineering — *Sept 2015 – Dec 2021*

Thesis advisers: Dr. Stephen Quake, Dr. John Boothroyd

Reed College — Portland, Oregon, U.S.A.

B.A. in Biology — *Sept 2010 – May 2014*

Thesis adviser: Dr. Jay Mellies

Publications

1. **Yuan Xue***, Yuhong Ning*, *et al.*, Wayne Volkmuth, Samuel Levy‡. 5-hydroxymethylcytosine analysis reveals stable epigenetic changes in tumor tissue that enable cancer detection in cell-free DNA. **Communications Biology (2025).**

2. Shimul Chowdhury, Michael Kesling, Micah Collins, Vanessa Lopez, **Yuan Xue**, Glenn Oliveira, Verena Friedl, Anna Bergamaschi, David Haan, Wayne Volkmuth, Samuel Levy‡. Analytical Validation of an Early Detection Pancreatic Cancer Test Using 5-Hydroxymethylation Signatures. **The Journal of Molecular Diagnostics (2024).**

3. Dania Nanes Sarfati, **Yuan Xue**, Eun Sun Song, Ashley Byrne, Daniel Le, Spyros Darmanis, Stephen R. Quake, Adrien Burlacot, James Sikes‡, Bo Wang‡. Coordinated wound responses in a regenerative animal-algal photosymbiotic metaorganism. **Nature Communications (2024).**

4. Trung Pham‡*, **Yuan Xue***, Susan Brewer, Kenneth E. Bernstein, Stephen R. Quake‡, Denise Monack‡. Single-cell profiling reveals functional diversity of granuloma macrophages during persistent Salmonella infection. **Science Advances (2023)**. bioRxiv preprint

5. **Yuan Xue**, Ido Braslavsky, Stephen R. Quake. Temperature effect on DNA polymerase fidelity. **Journal of Biological Chemistry (2021)**. bioRxiv preprint

6. Pengyang Li, Dania Nanes Sarfati*, **Yuan Xue***, Xi Yu, Alexander Tarashansky, Stephen R. Quake, Bo Wang. Single-cell analysis of Schistosoma mansoni reveals a conserved genetic program controlling germline stem cell fate. **Nature Communications (2020)**. bioRxiv preprint

7. Suchita Rastogi, **Yuan Xue**, Stephen R. Quake‡, John Boothroyd‡. Differential Impacts on Host Transcription by ROP and GRA Effectors from the Intracellular Parasite Toxoplasma gondii. **mBio (2020)**. bioRxiv preprint

8. **Yuan Xue**, Terence Theisen, Suchi Rastogi, Abel Ferrel, Stephen R. Quake‡, John Boothroyd‡. A single-parasite transcriptional atlas of Toxoplasma gondii reveals novel control of antigen expression. **eLife (2020)**. bioRxiv preprint

9. Alexander Tarashansky, **Yuan Xue**, Pengyang Li, Stephen R. Quake, Bo Wang. Self-assembling Manifolds in Single-cell RNA Sequencing Data. **Elife (2019)**. bioRxiv preprint

10. **The Tabula Muris Consortium**, Stephen R. Quake, Tony Wyss-Coray, Spyros Darmanis. Single-cell transcriptomics of 20 mouse organs creates a Tabula Muris. **Nature (2018)**. bioRxiv preprint

11. **Yuan Xue**, Jossef Osborn, Anand Panchal, Jay L. Mellies. The RpoE stress response pathway mediates reduction of enteropathogenic Escherichia coli virulence by zinc. **Applied and Environmental Microbiology (2015)**.

12. Jing Zhou, Shi-Hao Tan, Valerie Nicolas, Chantal Bauvy, Nai-Di Yang, Jianbin Zhang, **Yuan Xue**, Patrice Codogno, Han-Ming Shen. Activation of lysosomal function in the course of autophagy via mTORC1 suppression and autophagosome-lysosome fusion. **Cell Research (2013)**.

***equal contributions; ‡corresponding authors**

Conferences & Presentations

Poster presentation — CSHL Biological Data Science
5-hydroxymethylcytosine analysis reveals stable epigenomic changes in tumor tissue that enable cancer detection in cell-free DNA — *2024*

Oral presentation — ESMO Congress
5-Hydroxymethycytosine analysis reveals stable epigenetic changes in tumor tissue that enable cfDNA cancer predictions — *2022*

Invited talk — National University of Singapore
Building a single-cell atlas of Toxoplasma interactome — *2019*

Invited talk — Cell Symposia Single Cells: From Technology to Biology
Building a single-cell atlas of Toxoplasma interactome — *2019*

Poster presentation Stanford Bioengineering department retreat
Single-cell co-transcriptomic measurement resolves parasitic life cycle and host interactions *2018*

Invited talk Stanford Microbiology & Immunology department retreat
Building a single-cell atlas of Toxoplasma interactome *2018*

Poster presentation Stanford Bioengineering department retreat
Cool biochemistry measured with a hot tool *2017*

Poster presentation Gordon Research Conference: Nucleic Acids
Temperature adaptation and polymerase fidelity *2017*

Poster presentation Gordon Research Conference: Microbial Toxins & Pathogenicity
Molecular Mechanism of Zinc Disruption of Enteropathogenic Escherichia coli Pathogenesis *2014*

Awards & Honors

Bio-X Travel Award
Stanford Bio-X program *2019*

Stanford Interdisciplinary Graduate Fellowship
Awarded to eleven students to support their doctoral research with an interdisciplinary scope for three years. *2018*

Postbaccalaureate Research Fellowship
Reed College, Biology Department *2014*

Summer Experience Research Award
Reed College *2013*

Gold Team Medal
MIT iGEM Competition *2009*

Professional Service

Teaching Assistant Stanford University
Microfluidic Device Laboratory (BioE301D) *2018*

Teaching Assistant Stanford University
Introduction to Bioengineering (BioE80) *2017*

Teaching Assistant Reed College
Microbiology *2014*

Academic Tutor Reed College
Biology, Chemistry *2011 – 2014*

Skills

Languages: English (native), Cantonese (native), Mandarin (native), Japanese (conversational)

Technology Stack: AWS, Nextflow, Docker, Python, R, SQL, PyTorch, scikit-learn, DBT, Snakemake, Snowflake, Slurm

Bioinformatics: BWA, GATK, samtools, bedtools, Picard, edgeR/DESeq2, glmnet, MACS2, IchorCNA, GSEA, Scanpy/Seurat

Domains: liquid biopsy, cfDNA fragmentomics, epigenomics, single-cell genomics, cancer diagnostics, biomarker discovery, analytical validation, statistical modeling, deep learning